

Project Description (Ambulance Services)

Balancing an employee's workload is always challenging and a sensitive topic in the workplace. With the ever desire to improve work-life balance and make the workplace environment more conducive and attractive (Guest. 2002), companies are often looking for tools that can ease their workload management as work-life balance is a cycle (Lupu & Ruiz-Castro, 2021).

As a result, your client has decided that a workload management system on the Web would be an informative way to have an overview of the company's manpower strengthen at any time and get more informative availabilities of employees and engagement.

The system is expected to provide an interactive and visual way for the employee to see their job assignments, working hours engaged/assigned, and for employees to inform the system of their availabilities. Employees should be able to indicate to the company any assigned jobs they cannot fulfil ahead of time. To assist with the workload planning, employees should be able to indicate their availabilities up to one month earlier.

The administrative staff (usually the manager) should be able to visualise the manpower availability at any time up to one month earlier. Hence, the system is expected to provide an interactive and visual way for the administrative staff to visualise manpower availabilities, job assignments, and allocate jobs. Individual employee workload and availabilities should be able to be visualised at a glance.

The Company

In the ambulance team of a major hospital, work allocation is assigned weekly every Monday. The workload allocation planning will start every Thursday of the week. Hence, all employee's availabilities must be informed in the system every Wednesday to be considered in the planning. If employees miss the weekly deadline, requests would be dealt with on a case-by-case basis.

The weekly roster is dependent on the team's availability. Currently, there are ten ambulances. Each ambulance is made up of one driver, which can be double up as a care assistant, one care assistant, and two paramedics. This project will use the ambulance services roles from the following URL: <https://www.healthcareers.nhs.uk/explore-roles/ambulance-service-team/roles-ambulance-service>

The manager will make their assignments such that there should be at least six ambulances on service at any time. The team consist of 17 drivers, 20 care assistants, and 36 paramedics. Each shift is 12 hours long and all members of the team should have at least 12 hours break within shifts. It takes up to 15 minutes to clean the ambulance between service. Ambulances will take about 1 hour to service, clean, and refuel between shifts. Therefore, an ambulance can only be in service at most for 11 hours in a 12-hour shift.

Whenever a staff is not on duty, the staff will be on standby to stand-in for cases on when another staff is on sick leave.

Initial Requirements

We have sat down and brainstormed what we want to see in the Web application. As with any annoying client, we may be unclear, unreasonable and contradictory, amongst other common traits you may know of clients. Nevertheless, the following is our best attempt at listing what we want currently:

1. The app should be Web-based in a language of your choosing
2. The manager should be able to visualise the staff workload immediately on the landing page
3. The manager should be able to allocate jobs to staff for one week at a time
4. The manager should be able to view up to three staff availability and any relevant information to make the job assignment easier on the job allocation page
5. When displaying the staff availability, the workload assigned, staff's job preference, staff's location at a particular date, and availabilities for the week should be shown
6. On the manager's landing page, the top three staff with the lowest workload should be shown, and highlight all staff over 40 hours of jobs allocated
7. The manager should have a page that contains information of ambulances utilisation rate, and which ambulance(s) and slot(s) are requiring manpower
8. Staff should be able to view their weekly job assignments and overall workload for the month on their landing page
9. Staff can add and edit their availabilities up to 5 weeks ahead of time.
10. Staff can indicate their job preference for the week
11. Staff can reject jobs assigned to them, but they will be warned to discuss the jobs with their manager before proceeding with the rejection
12. The company's IT administrators will oversee adding new staff and managers to the system

References

Guest, D. E. (2002) 'Perspectives on the Study of Work-life Balance', *Social Science Information*, 41(2), pp. 255–279. doi: 10.1177/0539018402041002005.

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Lupu, I. and Ruiz-Castro, M. (2021) "Work-Life Balance Is a Cycle, Not an Achievement". Last accessed 22 July 2022. <https://hbr.org/2021/01/work-life-balance-is-a-cycle-not-an-achievement>